

STUDY DESIGN

STAGE 1: DATA PREPARATION

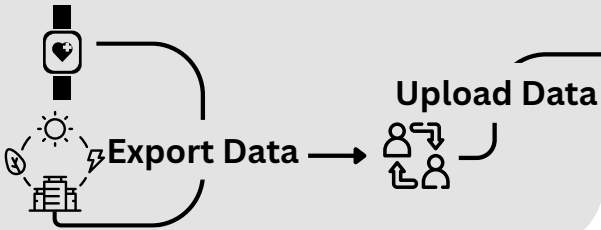
Data Collection

SURVEY:

- 17 households
 - Demographics
 - Utility usage
 - Fitbit activity

Data Processing

- Data Assessment
- Data Cleaning



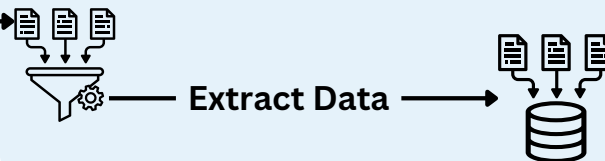
STAGE 2: ENGAGING FEATURES

Input Target Features

- Create Target Feature:
 - +24h (1-day ahead)
 - +168h (1-week ahead)
 - +720h (1-month ahead)
- Create New Features (Lag)

Feature Selection Process

- Features Correlation with the Target:
 - correlation > 0.05
- Select top most important features



STAGE 3: FORECASTING MODELS

Validation Approach

Walk-forward forecasting

Compare Performance Before & After Integrating Environmental & Behavioral Features

Models (XGBoost, Ridge, LSTM, CNN-LSTM)

Evaluation Metric

(MAPE%) Evaluated the Accuracy of Forecasting

Analyze Data

